Lab 5 — 2D Isometric Drawing

AutoCAD Masterclass (C391) · Topic 02: Technical Drawing with AutoCAD

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Objective

LO2 — produce 2D and 3D technical drawings using AutoCAD

Goal

Use ISODRAFT to create a 2D isometric representation of the 3D object supplied in iso-object.dwg, switching isoplanes and drawing isocircles for the round features.

You'll produce

A flat isometric drawing of the 3D object — edges on the three isoplanes plus isocircle holes.

Commands / tools: ISODRAFT, isoplanes (F5), ELLIPSE Isocircle, polar tracking

Files in this lab

- iso-object.dwg
- iso-object-sheet.pdf
- iso-object.png

Step-by-step

1. Open the object sheet iso-object-sheet.pdf and identify its three faces
2. Turn on isometric drafting from the status bar and set the Top isoplane

   ```
   ISODRAFT → Isoplane Top (F5)
   ```

3. Draw the top face edges along the 30°/150° isometric axes with polar tracking

   ```
   L with polar 30°/150°
   ```

4. Switch to the Right and Left isoplanes and complete the vertical faces

   ```
   F5 → Right · F5 → Left
   ```

5. Add the round features as isocircles on the correct isoplane

   ```
   EL → I (Isocircle) → centre → radius
   ```

6. Compare your isometric with the 3D model in iso-object.dwg and save

   ```
   SAVEAS iso-mine.dwg
   ```

Check your work

Each face lies on the correct isoplane (edges at 30°/90°/150°), the isocircles read as ellipses aligned to their faces, and the drawing matches the supplied object's proportions.

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